Hypothesis Test Report

Do Electronics orders generate significantly higher sales than Fashion orders?

1. Business question and hypotheses

H0: Average order value (Electronics) = Average order value (Fashion).
H1: Average order value (Electronics) > Average order value (Fashion) (one-sided).
Significance level α = 0.05. Unit of analysis: one order (Total_Sales).

2. Data

	Electronics	Fashion
Orders (n)	354	156
Mean order value	₹143.4K	₹127.2K
Median order value	₹114.5K	₹101.6K
95% CI for mean	₹131.1K to ₹155.7K	₹110.5K to ₹143.8K
Skewness	0.98	1.20

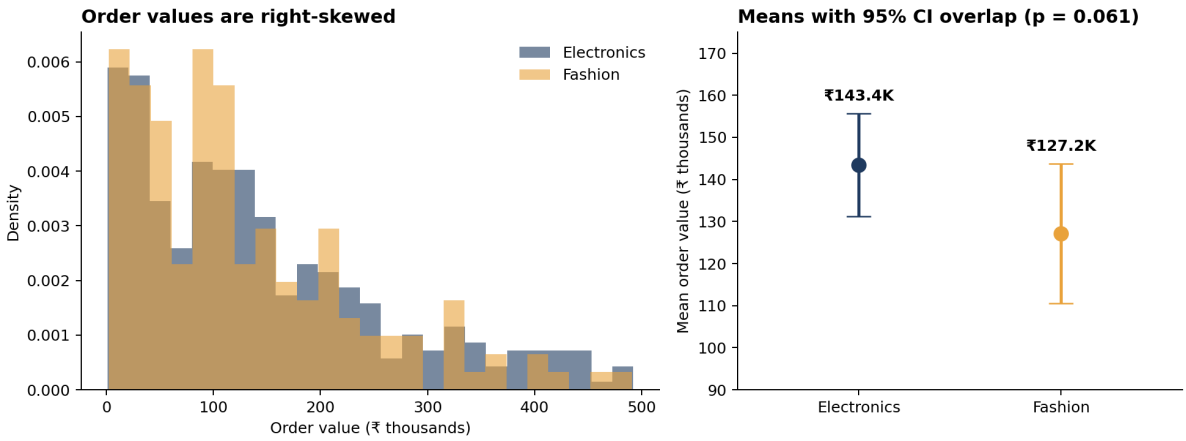
3. Choice of test

Two independent groups and a numeric outcome call for a two-sample t-test. Group sizes differ and variances are not clearly equal (Levene p = 0.093), so the **Welch t-test** is used. Order values are right-skewed, so a Mann–Whitney test and a Welch test on log-transformed values are run as robustness checks; with n above 150 per group the t-test remains reasonable by the central limit theorem.

4. Results

Test	Statistic	p-value	Reading
Welch t-test (primary, one-sided)	t = 1.55, df = 329	0.061	Not below 0.05
Mann–Whitney U (one-sided)	rank-based	0.116	Not significant
Welch on log(sales) (one-sided)	log scale	0.213	Not significant
ANOVA, all 5 categories	F = 0.72	0.580	No category differs
Kruskal–Wallis, all 5 categories	rank-based	0.735	No category differs

Effect size and intervals	Value
Mean difference (Electronics − Fashion)	₹16.3K (12.8% of Fashion mean)
95% CI for difference (Welch)	−₹4.3K to ₹36.9K
95% CI for difference (bootstrap, 10,000 resamples)	−₹4.5K to ₹36.9K
One-sided 95% lower bound	−₹1.0K
Cohen's d	0.14 (small)
Bootstrap share of resamples with Electronics higher	94.0%
Orders per group for 80% power at this effect size	606



5. Decision and interpretation

Fail to reject H0. With p = 0.061 (above α = 0.05), the data do not provide strong enough evidence that Electronics orders are larger than Fashion orders. The observed gap of ₹16.3K is plausible under random variation: the 95% confidence interval runs from −₹4.3K to ₹36.9K and includes zero. The result would be called significant at a 10% level, but that is a looser standard and was not our pre-set threshold. Failing to reject H0 is not proof that the groups are equal; the test may simply lack power for an effect this small.

Business conclusion: Electronics' revenue lead comes mainly from selling more orders (two products, 354 orders) rather than larger orders. Do not base pricing or inventory decisions on an assumed order-size premium without more data.

6. Limitations

(a) About 606 orders per category are needed to detect an effect this size reliably. (b) The data are observational, from one year. (c) Average unit price is nearly identical across all six products (about ₹24.7K to ₹26.4K, including Rice and Laptop), which is unusual for real retail; the data source and price field should be confirmed. (d) Dataset currency is assumed to be INR.